

CENTRAL BOARD EXAMINATION - 2023-2024

CLASS 10 — MATHEMATICS

Time Allowed: 3 Hours

Maximum Marks: 80

General Instructions:

1. This question paper contains 10 questions divided into Section A and Section B.
2. Section A contains 5 Multiple Choice Questions carrying 1 mark each.
3. Section B contains short & long answer questions carrying 3 to 5 marks each.
4. All questions are compulsory. Read the diagrams carefully where provided.

SECTION A (Multiple Choice Questions)

Q1. What is the SI unit of electric current?

- (a) Volt (b) Ampere (c) Ohm (d) Joule

Q2. Which of the following is a balanced chemical equation?

- (a) $\text{H}_2 + \text{O}_2 \rightarrow \text{H}_2\text{O}$ (b) $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$ (c) $\text{H}_2 + 2\text{O}_2 \rightarrow \text{H}_2\text{O}$

Q3. The focal length of a spherical mirror is 20 cm. Its radius of curvature is:

- (a) 10 cm (b) 20 cm (c) 40 cm (d) 5 cm

Q4. Which gas is evolved when dilute hydrochloric acid reacts with zinc granules?

- (a) Oxygen (b) Hydrogen (c) Carbon dioxide (d) Nitrogen

Q5. The speed of light in vacuum is approximately:

- (a) 3×10^8 m/s (b) 3×10^5 m/s (c) 1.5×10^8 m/s (d) 3×10^6 m/s

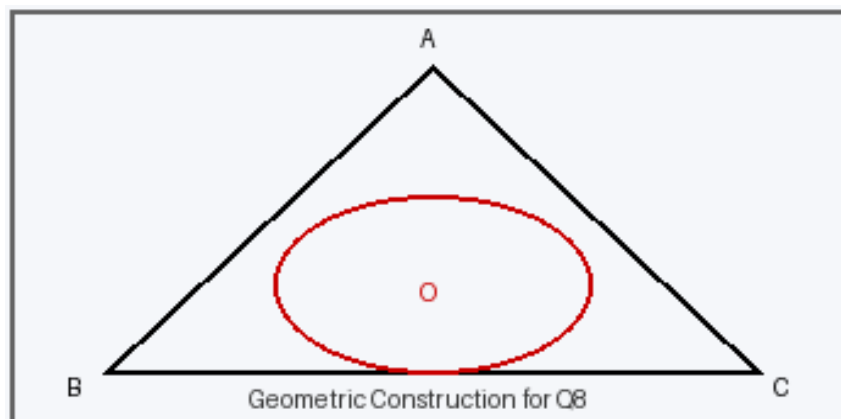
SECTION B (Short & Long Answer Questions)

Q6. State Ohm's Law and write its mathematical formula.

- (a) Define resistance of a conductor.
(b) What factors affect the resistance of a wire?

Q7. Differentiate between autotrophic and heterotrophic nutrition with examples.

Q8. In the figure given below, a circle is inscribed in triangle ABC:



- (a) Prove that the lengths of tangents drawn from an external point to a circle are equal.
(b) If $AB = 12$ cm, $BC = 8$ cm, and $AC = 10$ cm, find the lengths of the tangent segments.

Q9. Find the quadratic polynomial whose zeroes are 3 and -4.

Q10. Explain the process of photosynthesis with a balanced chemical equation.